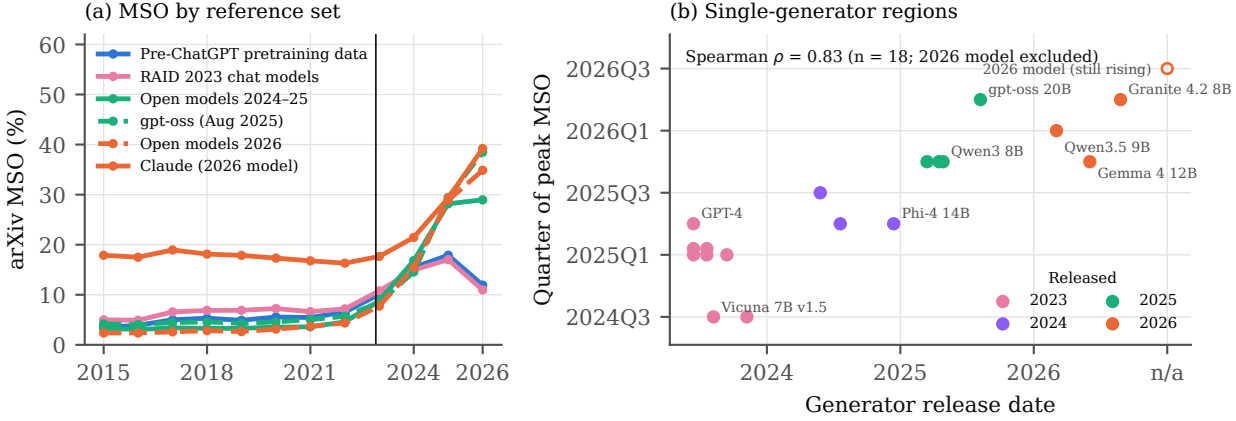
(a) MSO by reference set
arXiv MSO (%)
60
50
40
30
20
10
0
2015
2018
2021
2024
2026
Pre-ChatGPT pretraining data
RAID 2023 chat models
Open models 2024–25
gpt-oss (Aug 2025)
Open models 2026
Claude (2026 model)
(b) Single-generator regions
Spearman ρ = 0.83 (n = 18; 2026 model excluded)
Quarter of peak MSO
2026Q3
2026Q1
2025Q3
2025Q1
2024Q3
2026 model (still rising)
gpt-oss 20B
Granite 4.2 8B
Qwen3.5 9B
Qwen3 8B
Gemma 4 12B
GPT-4
Phi-4 14B
Vicuna 7B v1.5
Released
2023
2024
2025
2026
2024
2025
2026
n/a
Generator release date